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1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 6.1 Decision Tree Classifier
5  Machine Learning for Engineering Problem Solving
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7  """
8
9  import IPython as IP
10 IP.get_ipython().run_line_magic('reset', '-sf')
11
12 import numpy as np
13 import matplotlib.pyplot as plt
14 import sklearn as sk
15 from sklearn.datasets import load_iris
16 from sklearn.tree import export_graphviz
17 import graphviz as graphviz
18
19 plt.close('all')
20
21
22 %% Load your data
23
24 # We will use the Iris data set. This dataset was created by biologist Ronald
25 # Fisher in his 1936 paper "The use of multiple measurements in taxonomic
26 # problems" as an example of linear discriminant analysis
27 iris = sk.datasets.load_iris()
28
29 # for simplicity, extract some of the data sets
30 X = iris['data'] # this contains the length of the petals and sepals
31 Y = iris['target'] # contains what type of flower it is
32 Y_names = iris['target_names'] # contains the name that aligns with the type of the
    flower
33 feature_names = iris['feature_names'] # the names of the features
34
35 %% Build the model
36
37 # train the decision tree
38 tree_clf = sk.tree.DecisionTreeClassifier(max_depth=3)
39 X_petal = X[:,2:]
40 tree_clf.fit(X_petal, Y)
41
42
43 %% Visualize the decision tree
44
45 #create the export file for graphviz and export it. The file is exported as a
46 #.DOT file and can be viewed in an online viewer
47 https://dreampuf.github.io/GraphvizOnline/
48 export_graphviz(
49     tree_clf,
50     out_file="tree_clf.dot",
51     feature_names=feature_names[2:],
52     class_names=Y_names,
53     rounded=True,
54     filled=True
55 )
56
57 # We can load the file back in
58 s = graphviz.Source.from_file('tree_clf.dot')
59
60 # look at what is inside it. Also, just typing s in the console will display the image
61 print(s)
62
63 # export the image to a jpg
64 s.render('tree_clf', format='jpg', view=True)
65
66 %% Predict the class for any petal size

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66 size = [[7, 2.5]]
67 print(tree_clf.predict_proba(size))
68 print(iris.target_names)
69
70
71
72 # plot the new data point over the Iris dataset
73 plt.figure()
74 plt.grid(True)
75 plt.scatter(X[Y==0,2],X[Y==0,3],marker='o',label=Y_names[0],zorder=2)
76 plt.scatter(X[Y==1,2],X[Y==1,3],marker='s',label=Y_names[1],zorder=2)
77 plt.scatter(X[Y==2,2],X[Y==2,3],marker='d',label=Y_names[2],zorder=2)
78 plt.scatter(size[0][0],size[0][1],s=300,marker='*',label='new data point',zorder=2)
79 plt.xlabel(feature_names[2])
80 plt.ylabel(feature_names[3])
81 plt.legend(framealpha=1)
82 plt.tight_layout()
83
84
85
86
87
```